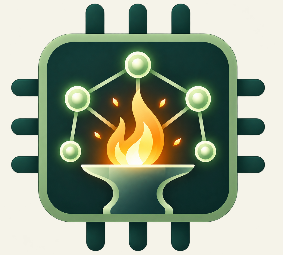


DataForge 2026

MemoryForge

Associative Memory & Fast Weights

PS-1 — Pathway Track



Team BitWise

MEMBER

Ved Patel

Deepshikha Rani

ASSIGNED RESPONSIBILITIES

ML engineering, application development,
evaluation

Research, technical writing, presentation

Learning an association without retraining

The problem

A changed prediction does not show what the model learned. MemoryForge exposes the state that changed.

Encoder parameter delta

0

Fast-memory delta

1.73205 > 0

The screenshot shows the MemoryForge interactive learning lab interface. On the left, a sidebar contains the MemoryForge logo and a 'YOUR LEARNING PATH' section with options: Guided lab (selected), Playground, Research & evidence, and 60-second check. Below this, it states 'ENCODER - FROZEN' and 'Episode 1000'. The main area is titled 'INTERACTIVE LEARNING LAB / FAST-WEIGHT ASSOCIATIVE MEMORY' and features a header 'New associations. Same neural weights.' with a sub-header 'Can an AI learn a new association without changing its neural-network weights? Teach this memory, query it, then clear it to find out.' Below the header are three tabs: '01 - Represent' (A frozen encoder turns a digit into a vector.), '02 - Associate' (Demonstrations write new symbol associations.), and '03 - Retrieve' (Unseen images read the temporary memory.). The '02 - Associate' tab is active, showing a 'Teach demonstrations' button. Below the tabs are three metrics: 'Encoder parameter delta' (0, Frozen - exact tensor equality verified), 'Fast-memory delta' (1.73205, Measured from the empty retrieval matrix), and 'Actual memory writes' (3, 1 clean demos/class - 0 conflicting writes). The '01 / Teach an association' section shows three digit images with their associations: Digit 1 → ALPHA, Digit 4 → GAMMA, and Digit 8 → BETA. The '02 / Retrieve a new association' section shows a digit image (Digit 1, sample 485) with a prediction of ALPHA and a ground truth of ALPHA, indicating a 'Correct association'.

Working CPU prototype. Seed 1000 preset: 3 writes and 23/30 fixed queries correct.

The write rule and the retrieval rule

WRITE A DEMONSTRATION

$$\mathbf{S} \leftarrow \mathbf{S} + \mathbf{v} \mathbf{k}^T$$

$$\mathbf{c} \leftarrow \mathbf{c} + \mathbf{v}$$

$$\mathbf{M}[i] = \mathbf{S}[i] / \mathbf{c}[i]$$

Zero row when $\mathbf{c}[i] = 0$

READ A QUERY

$$\text{scores} = \mathbf{M} \mathbf{q}$$

Empty memory abstains. Scores are similarities, not calibrated probabilities.

What the symbols mean

$\mathbf{k}$ and $\mathbf{q}$: unit 16-D embeddings

$\mathbf{v}$: one-hot temporary label

$\mathbf{S}$: key sums, $\mathbf{c}$: write counts

What changes

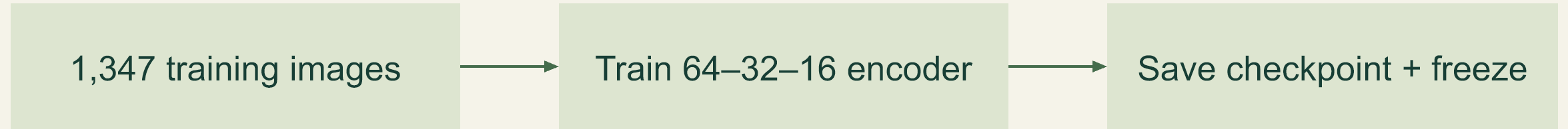
$\mathbf{S}$ and $\mathbf{c}$ change when taught.

$\mathbf{M}$ holds one mean per label.

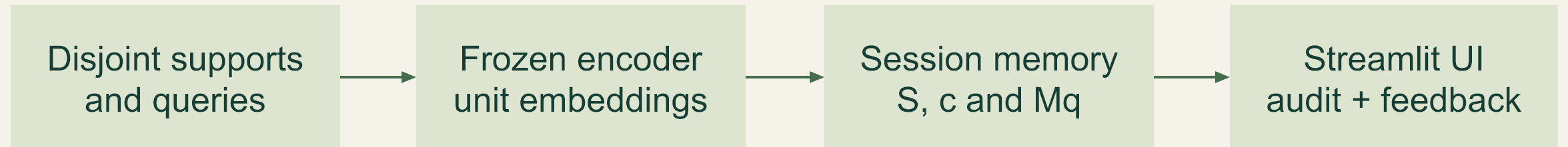
Encoder tensors remain fixed.

Architecture of the working prototype

Offline preparation



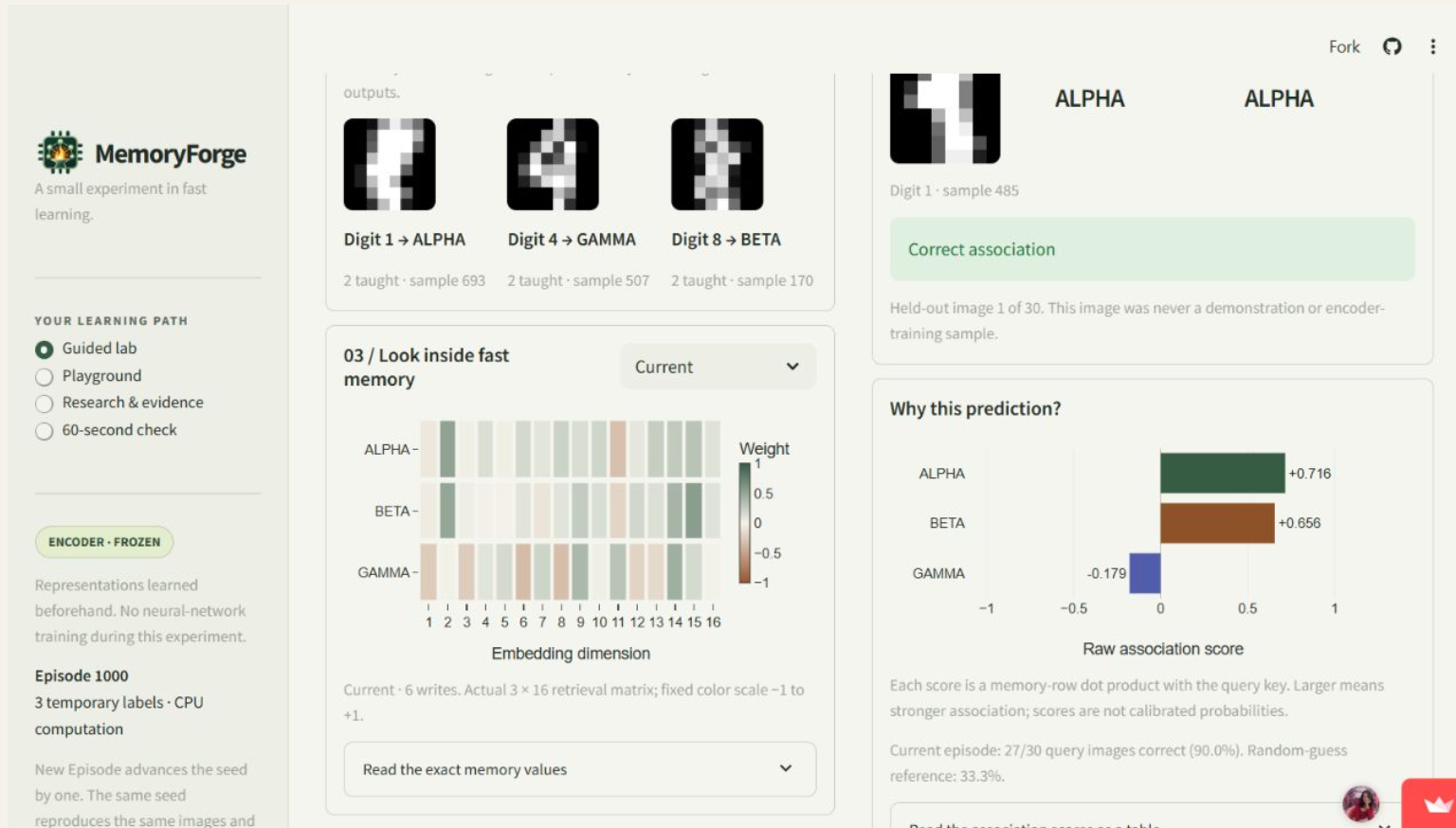
Runtime on CPU: 450 held-out images



Python 3.12 / PyTorch / scikit-learn / NumPy / Streamlit / Plotly

No model API or database. Session reload clears memory. Detailed DFD: docs/DATA_FLOW.md.

A learner can inspect each state change



1 Teach

Add one support per class.
S, c and M update.

2 Test Query

Read M_q on a held-out image.
Memory stays unchanged.

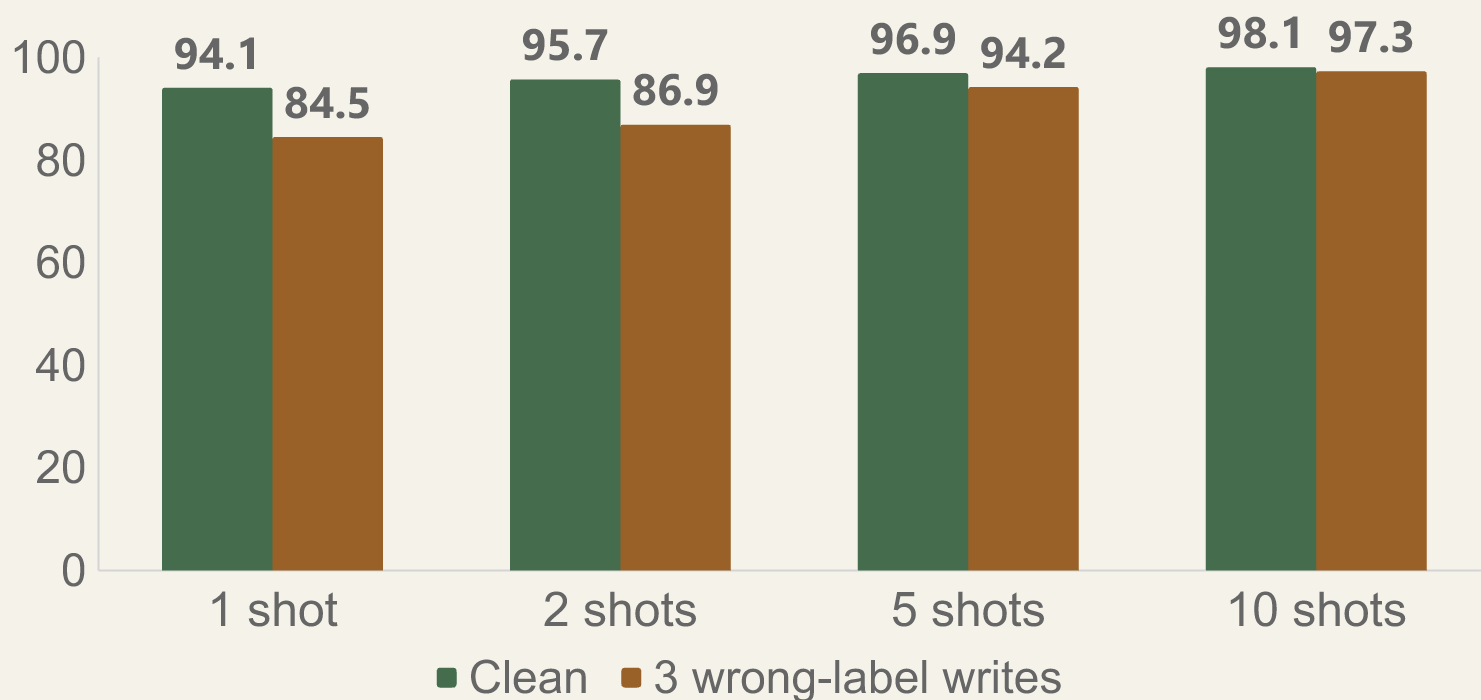
3 Clear Memory

Zero the memory.
The predictor abstains.

Guided lab, Playground, sourced research, and a 60-second concept check are implemented.

More support reduces average conflict damage

Mean query accuracy (%)



50 fixed seeds

650 conditions
19,500 query outcomes
Encoder delta = 0
throughout

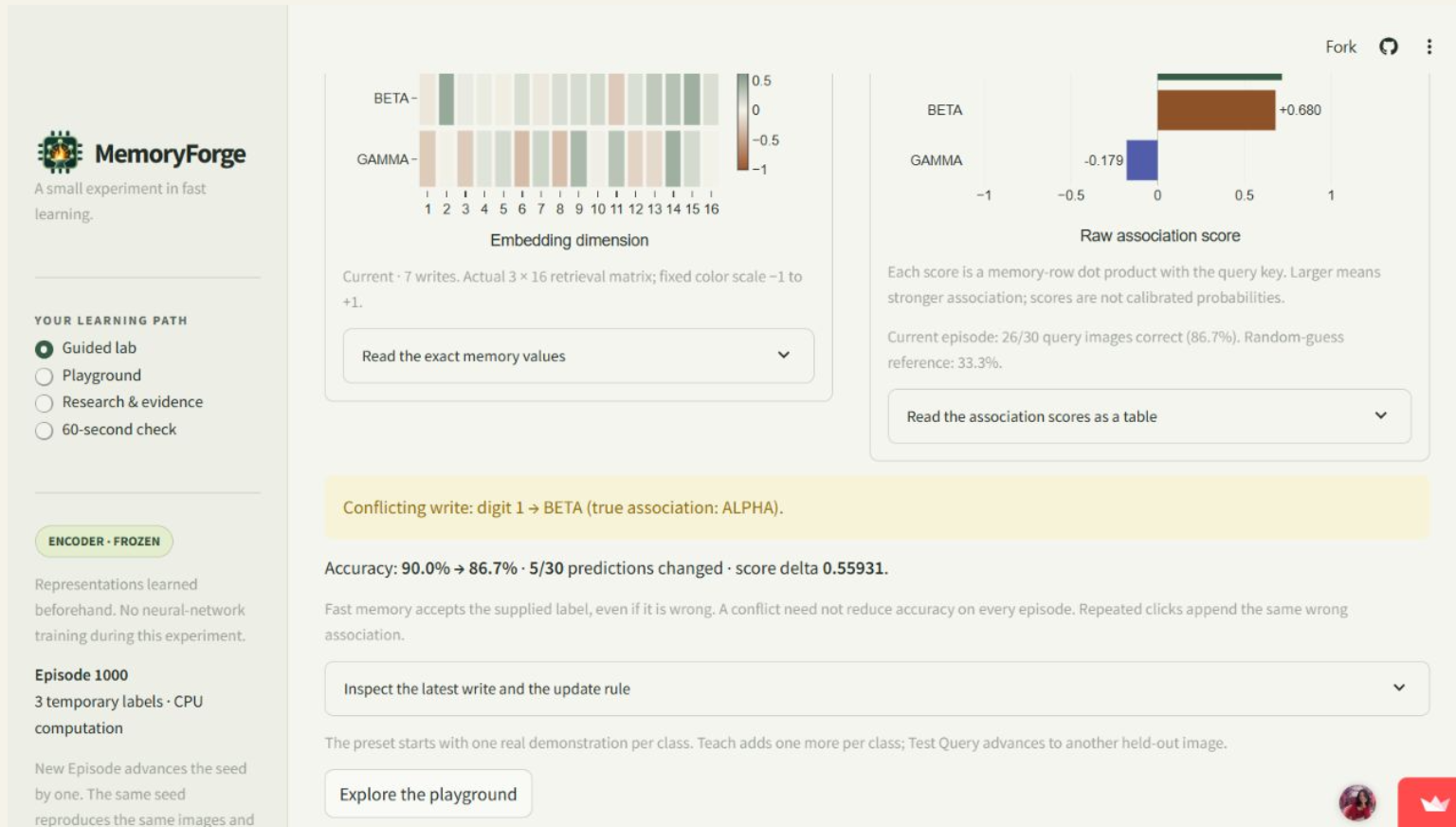
A limit worth testing

More shots can hurt one seed.

No novel-category or concurrent-load evaluation.

Developer evaluation. Shared images and unequal corruption fractions limit the comparison.

A wrong label changes the answer



27/30 → 26/30

correct after the conflict

5 predictions change

The mean for BETA now includes an ALPHA key.

The lesson

A writable memory also accepts contradictory input.

Seed 1000, the same 30 queries. One mislabeled write; the original ground truth is preserved.

The research connection is about changing state

BDH [1]

Fixed connections coexist with Hebbian synaptic state.
This motivates the teaching distinction.

BDH-CQ [2]

Context memory feeds an iterative query workspace. Exact updates and dimensions remain proprietary.

Alternative update rules

DeltaNet [3]: retrieval-error correction.
Titans [4]: neural memory optimized at test time.

MemoryForge's boundary

Class-mean label memory only.
No BDH graph or CQ workspace.
No claim of general reasoning.

[1] 2509.26507v1 [2] 2608.09888v1 [3] 2406.06484v3 [4] 2501.00663v1

PS-1 / Pathway: a real state-changing experiment, inspectable evidence, and sourced BDH explanation.

What it can support, and what needs validation

Reuse and scale

An inspectable CPU teaching lab.

Core S+c state: 408 bytes.

Dense reads grow with labels

and embedding width.

Total app memory is larger.

Next experiments

Hold the noise fraction fixed.

Compare memory update rules.

Test unfamiliar categories.

Measure learner understanding

and concurrent-session behavior.

Sustainability option

Keep the teaching core open.

Explore paid workshops or managed classroom hosting.

Validate demand and support costs before offering a service.

Functional prototype today

Public CPU app, 124 tests passed, reproducible evidence.

Learner outcomes and production capacity remain unmeasured.

Proposals only. No current revenue, paying users, learning-outcome study, or production-readiness claim.

DataForge 2026

THANK YOU

MemoryForge / Team BitWise

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